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Dysfunctional State of T Cells or Exhaustion During Chronic Viral Infections and COVID-19: A Review

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Abstract

Coronavirus disease 2019 (COVID-19) continuously affecting the lives of millions of people. The virus is spread through the respiratory route to an uninfected person, causing mild-to-moderate respiratory disease-like symptoms that sometimes progress to severe form and can be fatal. When the host is infected with the virus, both innate and adaptive immunity comes into play. The effector T cells act as the master player of adaptive immune response in eradicating the virus from the system. But during cancer and chronic viral infections, the fate of an effector T cell is altered, and the T cell may enter a state of exhaustion, which is marked by loss of effector function, depleted proliferative capacity and cytotoxic effect accomplished by an increased expression of numerous inhibitory receptors such as programmed cell death protein 1 (PD-1), lymphocyte-activation protein 3 (LAG-3), and cytotoxic T lymphocyte-associated antigen 4 (CTLA-4) on their surface. Various other transcriptional and epigenetic changes take place inside the T cell when it enters into an exhausted state. Latest studies point toward the induction of an abnormal immune response such as lymphopenia, cytokine storm, and T cell exhaustion during SARS-CoV-2 (severe acute respiratory syndrome coronavirus 2) infection. This review sheds light on the dysfunctional state of T cells during chronic viral infection and COVID-19. Understanding the cause and the effect of T cell exhaustion observed during COVID-19 may help resolve new therapeutic potentials for treating chronic infections and other diseases.

Keywords: inflammation, T cell exhaustion, cytokines, COVID-19, virus

Introduction

CORONAVIRUS DISEASE 2019 (COVID-19) is a respiratory and contagious disease, which is caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) (17,54). This disease has emerged as a pandemic, leading to health emergency and causing major challenges to the health care system globally. The infection is associated with a fatality rate of 6.4%, posing a significant threat to humankind (19). The SARS-CoV-2 infection causes mild-to-moderate symptoms related to a respiratory illness that can be treated without any medical intervention. Still, in some cases, the symptoms can progress into a more severe form leading to death.

The probability of developing complications is higher in the elderly and in those suffering a medical condition, such as heart disease, diabetes, chronic respiratory infections, and cancer. COVID-19 is a communicable disease, which transmits primarily through the exposure to respiratory fluids contaminated with infectious virus. The exposure of SARS-CoV-2 may occur in three possible ways: (a) inhalation of respiratory droplets and aerosol particles; (b) deposition of respiratory droplets and particles on exposed mucous membranes in the mouth, nose, or eye; and (c) touching mucous membranes with hands that have been contaminated with virus.

During viral infection, the innate immune system acts as primary defense barrier against the incursive virus to

prevent the invasion or replication of the virus. The innate or non-specific arm of the immune system allows the pattern recognition receptors (PRRs) to recognize the viral specific components, such as viral nucleic acids (DNA or RNA), proteins, and other intermediate products. The interaction between PRRs and viral components stimulates the production and release of signaling molecules such as proinflammatory cytokines and type I interferons (IFNs α and β) from the infected cells, eliciting an immune response. PRRs play a crucial role in recognizing molecules usually associated with the invading pathogen leading to the generation of adaptive immune response (20). It majorly involves antigen-presenting cells, T and B lymphocytes. T cells orchestrate the adaptive immune response by identifying and eliminating the virus-infected cell, constituting cell-mediated immunity (5).

When an unprimed T cell comes in contact with an antigenic peptide, it differentiates into effector T cells, mediating the adaptive immune response. Whereas B lymphocytes or B cells elicit humoral immune response where they are primarily involved in the production of antigen-specific antibodies that neutralize the virus (16). The activated T cell population mediates the process of viral clearance, where the cytotoxic CD8⁺ T lymphocytes (CTLs) eliminate the virus by perforin and granzymes and cytokines such as IFN- γ (29). The helper CD4⁺ T cells help in polarizing the immune response, which is appropriate for the infection by assisting CTLs and B lymphocytes. T helper cells are further categorized based on the cytokines they secrete, T helper 1 (Th1) cell-mediated immune responses are predominantly driven by secretion of IFN- γ and interleukin (IL)-2, whereas IL-10 and IL-4 are dominated in T helper 2 responses, which control antibody-mediated immune response.

During the initial phase of infection, unprimed T cells recognize pathogen-derived molecules presented to the T cells via major histocompatibility complex leading to their activation, proliferation, and differentiation. These activated T cells then differentiate into functional T cells through transcriptional, epigenetic, and metabolic changes leading to antigen clearance. After the inflammatory response subsides and the antigen is eliminated, most of the effector T cells undergo programmed cell death. The leftover antigen-reactive T cells, called memory T cells, enter a state of quiescence that maintains long-lasting memory of the previous encounter. These memory T cells are crucial as they rapidly elicit an immune response when they re-encounter the same antigen. These effector functions are crucial to combat infections. But during cancers and chronic viral infections, the fate of these effector T cells is altered, and the T cell enters a state of exhaustion marked by an ineffective antiviral immune response.

The Loss of Effector Function and T Cell Exhaustion

A dysfunctional state of the T cell arises when the host is challenged with a persistent viral infection, which is known as T cell exhaustion. The cytotoxic and helper function of these cells becomes compromised, which leads to weakened immune response and persistent infection. The proof of exhaustion of CD8⁺ T cell was initially reported in animal model infected with lymphocytic choriomeningitis virus (LCMV) (32,42), where the cytotoxic CD8⁺ T cells were not

able to induce the production of cytokines and failed to clear the infection. The cells with this phenotype were characterized as exhausted T cells, and when compared with normal effector T cells, they exhibited altered transcriptional, metabolic, and epigenetic status along with increase in transcription of inhibitory receptors (IRs) and their expression (52).

The process of CD8⁺ T cell exhaustion (Fig. 1) takes place in a stepwise manner, beginning with the loss of production of IL-2 and gradual loss of proliferative capacity in CD8⁺ T cells. Followed by loss of ability to produce tumor necrosis factor-alpha (TNF- α) and diminished cytotoxic effect. Later on, complete loss of ability to produce IFN- γ and some chemokines is observed (58). This hierarchical process of T cell exhaustion leads to the loss of effector function, depleted proliferative capacity, impaired production of perforin (59), and suppression of cytotoxic T cell response, finally leading to cell death (43). The exhausted T cell phenotype is also observed in other viral infections, which persist in the host such as hepatitis B virus (HBV) (12), hepatitis C virus (HCV), human immunodeficiency virus (HIV) (46), and parasitic infection such as malaria (56), which involves continued exposure to the antigen.

Along with the exhaustion of CD8⁺ T cells, the CD4⁺ T helper cells also undergo loss of effector function in response to chronic viral infection (3). Accordingly, in LCMV infection, the CD4⁺ T cells depict an exhausted state similar to exhausted CD8⁺ T cells, suggesting that the CD4⁺ T cells also undergo exhaustion during chronic viral infections.

IRs and Exhausted T Cells

IRs have a significant role in controlling autoimmunity and self-tolerance in primed T cells, but increased and persistent expression of these receptors during chronic viral infection is correlated with the exhaustion of both cytotoxic and helper T cells (49).

Of all the IRs, the best characterized IR is programmed cell death protein 1 (PD-1) receptor, which is present on activated T cells and regulates the functioning of effector T cells. An enhanced and persistent expression of PD-1 is correlated with exhausted T cells during chronic viral infections. Inhibiting the pathway of PD-1 gene synthesis in LCMV-infected cells revived the exhausted CD8⁺ T cells and boosted the viral control function (1,2). Along with PD-1, exhausted T cells also express a high level of other IRs such as lymphocyte-activation protein 3 (LAG-3), T cell immunoglobulin and mucin-domain containing-3 (TIM-3), T cell immune receptor with Ig and ITIM domains (TIGIT), and cytotoxic T lymphocyte-associated antigen 4 (CTLA-4) (4,34,48,64).

LAG-3 plays an important role in regulating autoimmunity, tumor immunity, and adaptive immunity (23). Numerous evidence suggests that LAG-3 acts as an inhibitory co-receptor, and its expression is also enhanced on exhausted T cells in LCMV infection (41). During chronic HBV infection, the expression of LAG-3 increased on exhausted T cells, and on blocking the expression of LAG-3 led to the restoring of functioning of CD4⁺ T cells partially (10). However, the presence of single IR is insufficient to indicate an exhausted T cell as these cells synergistically co-express multiple IRs.

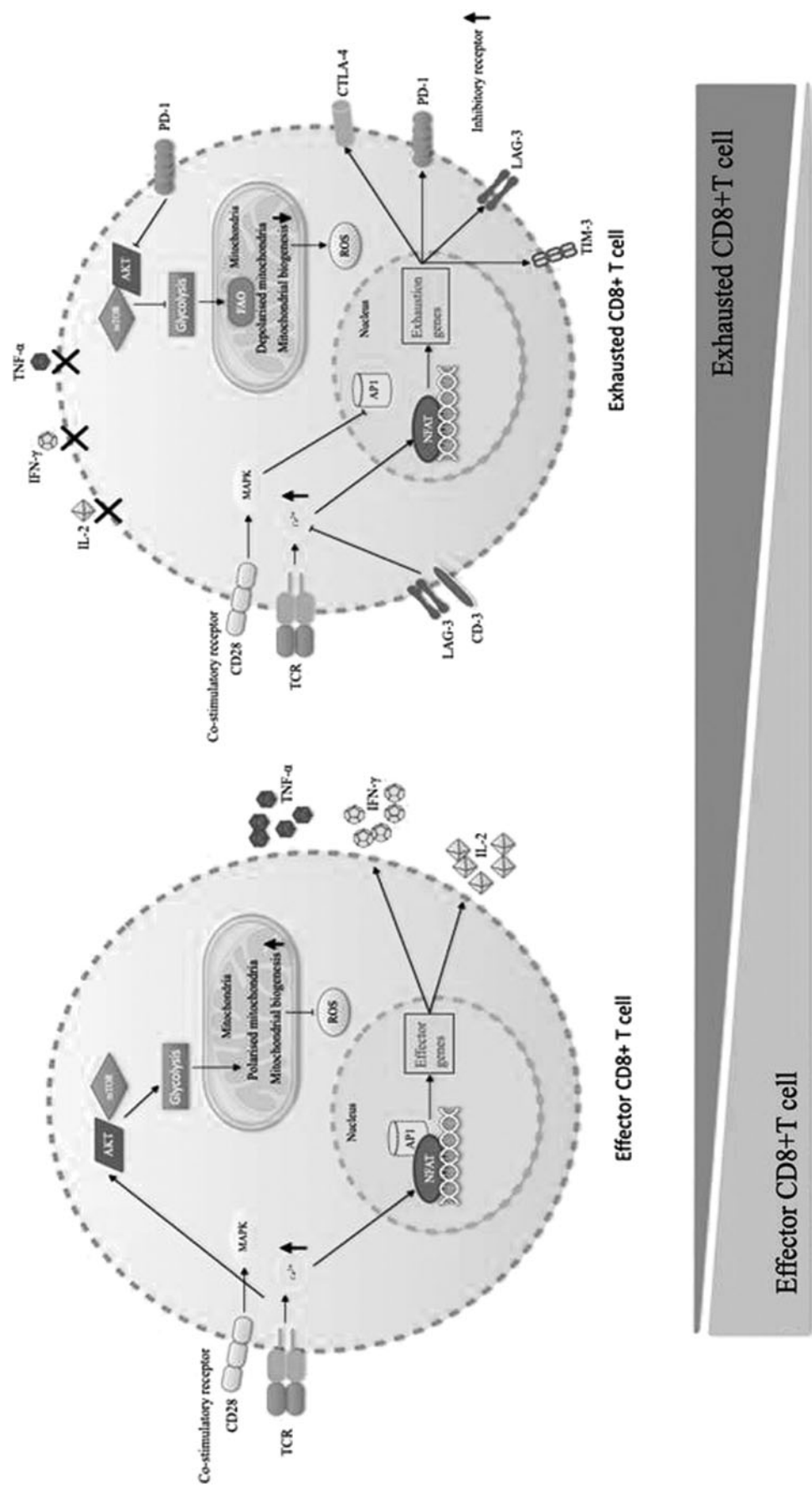


FIG. 1. Various markers associated with an effector and exhausted CD8⁺ T cell.

Inhibiting the expression of these IRs is a part of anti-cancer immunotherapies, which is known to improve the patient's outcomes significantly, revolutionizing the treatment of cancer. For example, inhibitors of PD-1 for the treatment of the various types of cancers are approved by the Food and Drug Administration (FDA). Some specific monoclonal antibodies produced against PD-1 and CTLA-4 are being used in cancer treatment (4,6,7).

Presently, extensive research is being carried out on exhausted T cells due to their clinical relevance in the development of therapeutic strategies against chronic infections (26,27). Several studies have pointed toward the notable importance of T cells in conferring protection against the severe form of COVID-19 disease (28,66). The presence of exhausted T cells is also reported in patients suffering with SARS-CoV-2 infection. Along with this, IRs such as TIM-3 and PD-1 are also detected on a subpopulation of T cells isolated from host infected with SARS-CoV-2 (65). In response to this, inhibitory monoclonal antibodies such as pembrolizumab and nivolumab against PD-1 and its ligand are under clinical trials to assess their role in preventing T cell exhaustion in COVID-19.

Role of Transcriptional and Epigenetic Factors in T Cell Exhaustion

In exhausted T cells, the transcription profile of certain genes is altered, including the genes involved in the pathways of cytokine signaling, costimulatory pathways, various T cell receptors (TCRs), energy metabolism, along with genes encoding multiple transcription factors and IRs (40,53). Studies on animal models have suggested the involvement of certain transcription factors such as nuclear factor of activated T cells (NFAT), thymocyte selection-associated high mobility group box protein (TOX), T-box transcription factor (TBX21 or T-bet), TCF1, B lymphocyte-induced maturation protein (BLIMP-1), and Eomesodermin (Eomes) in CD8⁺ T cell exhaustion (2,18,47). An example of transcriptional regulation of T cell exhaustion involves the transcription factor NFAT, which in the absence of co-stimulatory protein AP-1 induces the state of T cell exhaustion in both CD8⁺ and CD4⁺ T cells (22). It is also reported that the expression of a tran-

scriptional repressor Blimp-1 is substantially increased in exhausted CD8⁺ T cells, which relates to the enhanced expression of IRs.

It is reported in various studies that the epigenomic profile of an exhausted T cell is very different from the normal effector T cell. Chromatin regions of genes of IR PD-1 are present in an open configuration in exhausted T cells, where it remains inaccessible in normal effector T cells (35,36).

T Cell Exhaustion in Chronic Viral Infections

The effector functions of virus-specific CD8 T cell are exhausted during chronic infections (51). It appears to be a prominent feature of chronic viral infections or sustained viral antigen exposure to CD8 T cell. A summary of some of the exhaustion characteristics in different chronic viral infections such as HBV, HCV, LCMV, HIV, and Epstein-Barr virus is given in Table 1. Increased expression of PD-1, loss of effector function that includes loss of proliferation capability of T cells and decreased productions of cytokines such as IL-2, TNF- α , and IFN- γ , is the hallmark of T cell exhaustion in chronic viral infections.

T Cell Response in COVID-19

Latest studies have shed light on the immune response of patients suffering from SARS-CoV-2 infection. But the response of T cells (exhaustion or excessive activation) during SARS-CoV-2 infection remains to be investigated in depth (9,17,39,45). During SARS-CoV-2 infections, many studies have reported the successful induction of CD8⁺ T cell response (39). Some studies also reported the presence of COVID-19-specific CD8⁺ T cells in the blood of recovered patients, suggesting virus-specific T cell response as well as memory development (14,37). The activation and proliferation of effector T cell are marked by enhanced expression of the systemic chemokines and cytokines such as TNF- α , IFNs, CXCL8, IL-6, CXCL9, and CXCL10 during acute infections, which alters the T cell responses.

Lymphopenia is marked by an abnormal reduction in lymphocyte count, which is observed in patients suffering from SARS-CoV-2 infection, affecting natural killer (NK) cells, B cells, and CD4⁺ and CD8⁺ T cells. Lymphopenia

TABLE 1. T CELL EXHAUSTION IN CHRONIC VIRAL INFECTIONS

S. No.	Viruses causing exhaustion	Exhaustion characteristics	References
1	HBV	Increased PD-L1 expression on hepatocytes after hepatitis B infection	(33)
2	HCV	Dysfunctional antigen-specific CD8 T cells in chronic HBV infection	(11,44)
3	LCMV	Dysfunctional effector CD8 T cells in chronic HCV infection	(15,50)
		Dysfunctional effector function of CD8 T cells due to chronic infection and persistence viral antigen exposure	(58)
4	HIV	Hierarchical loss in the capacity to produce IL-2, TNF- α , and IFN- γ	(13)
		Impaired proliferative potential of T cells	(30)
		Increased PD-1 expression on virus-specific CTLs from HIV-infected patients	(8)
5	EBV	Increased PD-1 expression on CD8 T cells	(31)

CTL, cytotoxic T lymphocyte; EBV, Epstein-Barr virus; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; IFN- γ , interferon- γ ; IL-2, interleukin-2; LCMV, lymphocytic choriomeningitis virus; PD-L1, programmed death-ligand 1; TNF- α , tumor necrosis factor-alpha.

associated with other viral infections lasts for 2–4 days and recovers rapidly (25), whereas during SARS-CoV-2 infection, it is observed that lymphopenia is more persistent and is mostly associated with T cell lymphocytes. Prolonged lymphopenia-induced proliferation affects T cell activation and differentiation; however, direct correlation between lymphopenia and the process of T cell differentiation in COVID-19 patients is still under investigation. In patients suffering from chronic diseases, the expression of TNF- α , IL-10, and IL-6 is enhanced, which directly affects the population of T cells.

In a retrospective study conducted on 452 patients with COVID-19 at Tongji hospital, the authors reported a significant decrease in lymphocyte counts and increased levels of leukocytes that cause enhanced neutrophil–lymphocyte ratio, along with lower levels of monocytes, basophils, and eosinophils in severe patients. The level of biomarkers associated with infection and proinflammatory cytokines was also found to be elevated. Understanding how the reduction in lymphocyte count is related to disease progression can shed light on our understanding of the immune response during acute and chronic infection.

A large population of cytotoxic T cells is required to eradicate the infecting virus from the host system. Still, recent pieces of evidence indicated that the level of circulating T cells is reduced, and their function is impaired in patients suffering from COVID-19 and especially in those admitted to ICU. Different studies have reported that the level of both T helper cells, T cytotoxic cells, and T regulatory cells was below the normal range in severe cases of COVID-19 (6,38).

Initial studies related to adaptive immunity and SARS-CoV-2 infection have given an account of alteration in the state of CD8⁺ T cells. A single-seq analysis study revealed no significant evidence of exhaustion of CD8⁺ T cell in peripheral blood of patients infected with COVID-19, whereas the expression of several exhaustion markers was increased in CD4⁺ T cells, although these changes were insignificant (55). Interestingly, in another study, the expression of IRs such as PD-1, TIM-3, LAG-3, NKG2A, and CTLA-4 was increased in CD8⁺ T cells in patients suffering from SARS-CoV-2 infections, which points toward the exhausted state of T cells (61,62).

It was also observed that the presence of exhaustion markers on T cells in COVID-19 patients found enhanced expression of PD-1 and TIM-3 on CD8⁺ and CD4⁺ T cells in COVID-19 patients in comparison to healthy subjects, especially in patients admitted to ICU (55). Whereas the detailed examination of short conditional RNA in PBMCs revealed the enhanced exhaustion score of CD8⁺ T cells in patients suffering from severe SARS-CoV-2 infection in comparison to healthy individuals (14,60).

Another report suggests the decreased level of cytokine production in patients suffering from severe COVID-19 (63). Exhausted virus-specific CD4⁺ T cells are also known to express PD-1 at elevated levels, which are correlated with viral loads, disease progression, and CD4⁺ T cell reduction (61).

Latest study suggested that with a reduction in the proportion of regulatory T cells and an increment in the number of CCR6⁺ Th17 cells, a proinflammatory T cell response can be achieved (21). According to a study conducted on a

cohort of 68 patients suffering from COVID-19 (55 mild and 13 severe cases), NKG2A (CD94/NK group 2 member A), an IR known to induce exhaustion of NK cells during chronic infections, is identified to be principally expressed on CD8⁺ T cells and NK cells resulting in their functional and metabolic exhaustion. Also, a decrease in levels of CD107a (a degranulation marker), granzyme B, IFN- γ , and IL-2 was reported, indicating the loss of cytotoxic potential of T cells and NK cells. Fatal or severe COVID-19 cases were characterized by increased proportion, escalated activation, and sequestration of inflammatory CXCR4⁺ T cells in infected lungs. T cell dysfunction is positively associated with progression of disease and is a potential cause of aggravated pathological immune damage.

Xu *et al.* in his study reported a significant reduction in CD4⁺ as well as CD8⁺ T cells in peripheral blood along with their hyperactivation or exhaustion as evident by the enhanced levels of CD38 and HLA-DR (57). Additionally, the concentration of cytotoxic granules (perforin and granzyme), which are present in cytotoxic T lymphocytes, was comparatively higher. Moreover, an increase in the amount of CCR6⁺ Th17 in CD4⁺ T cells was observed (57).

During SARS-CoV-2 infection, the population of CD4⁺ T cells get activated, which are characterized by the presence of cell-specific markers such as Ki-67, CD25, HLA-DR, and CD38. Mathew *et al.* reported in a study that in patients hospitalized due to severe infection the share of cytotoxic T helper cells and follicular T helper cells was increased with reduction in the amount of virus-specific T regulatory cells, in comparison to nonhospitalized patients (24).

A recent study was conducted on two different cohorts, where the subjects of cohort 1 were those who had recovered from SARS-CoV-2 infection, whereas cohort 2 subjects were suffering from active SARS-CoV-2 infection. In the study, they investigated the functional state of T lymphocytes and found that COVID-19 strongly affects the expression of secretory immune checkpoint molecules, such as sCD25, Galectin-9, soluble Tim-3 (sTim-3), and sLAG-3. The expression of these molecules was usual after recovery from infection due to exaggerated antiviral response and counterregulation during infection period.

The count of activated T cells and the proportion of CD4⁺ to CD8⁺ were also found to be dysregulated in COVID-19 infection. Experiments on rhesus monkeys with depressed CD8⁺ induction upon artificial infection with SARS-CoV-2 revealed extinguished virus removal from the lungs (28). From the above-mentioned reports, it can be concluded that in SARS-CoV-2 infection, there is a significant decrease in functional T cell population characterized by T cell exhaustion. The crucial role of T cells and their exhaustion during persistent infection with SARS-CoV-2 remains to be studied in detail.

Conclusions

T cell exhaustion requires in-depth study to delineate the factors responsible for dysfunctional T cells. COVID-19 has brought out this issue in more vigorous way among researchers. There is a potential therapeutic application in subverting T cell exhaustion state for treatment of diseases, such as chronic viral infection, autoimmunity, and cancer.

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